HW #3 Solutions

1. Let $X = \text{span}\{S_1 \cup S_2\}$ and $Y = \text{span}\{S_1\} + \text{span}\{S_2\}$. We need to show that $X \subset Y$ and $Y \subset X$, because this is equivalent to $X = Y$. We denote the field by $\mathbb{F}$.

<u>Claim 1</u> $X \subset Y$.

<u>Proof</u> Let $x \in \text{span}\{S_1 \cup S_2\}$. Then $\exists$ $v_1, .., v_k \in S_1$ and $w_1, .., w_m \in S_2$, as well as $\alpha_1, .., \alpha_k \in \mathbb{F}$ and $\beta_1, .., \beta_m \in \mathbb{F}$ such that $x = \sum_{i=1}^{k} \alpha_i v_i + \sum_{i=1}^{m} \beta_i w_i$.

We observe ~~that~~, by definition of $\text{span}\{S_1\}$ and $\text{span}\{S_2\}$, that

$$\sum_{i=1}^{k} \alpha_i v_i \in \operatorname{span}\{S_1\} \text{ and } \sum_{i=1}^{m} \beta_i w_i \in \operatorname{span}\{S_2\}$$

Therefore, by the definition of the sum of two subspaces,

$$x = \sum_{i=1}^{k} \alpha_i v_i + \sum_{i=1}^{m} \beta_i w_i \in \operatorname{span}\{S_1\} + \operatorname{span}\{S_2\}.$$

$\square$

<u>Claim 2</u> : $\mathcal{Y} \subset \mathcal{X}$.

<u>Proof:</u> $y \in \mathcal{Y}$ means there exists

$y_1 \in \operatorname{span}\{S_1\}$ and $y_2 \in \operatorname{span}\{S_2\}$ such that $y = y_1 + y_2$.

$y_1 \in \operatorname{span}\{S_1\}$ implies there exist $\bar{\alpha}_1, \ldots, \bar{\alpha}_{\bar{k}} \in \mathcal{F}$ and $\bar{v}_1, \ldots, \bar{v}_{\bar{k}} \in S_1$ such that

$$y_1 = \sum_{i=1}^{\bar{k}} \bar{\alpha}_i \bar{v}_i .$$

Similarly, there exist $\bar{\beta}_1, .., \bar{\beta}_{\bar{m}} \in \mathcal{F}$ and $\bar{w}_1, .., \bar{w}_{\bar{m}} \in S_2$ such that

$$y_2 = \sum_{i=1}^{\bar{m}} \bar{\beta}_i \bar{w}_i .$$

Therefore $y = y_1 + y_2$ is a linear combination of elements of $S_1 \cup S_2$ and thus $y \in \operatorname{span}\{S_1 \cup S_2\} = X$.

$\square$

2.

(a) Linearly independent if, and only if, the only solution to

$$\alpha_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

is $\alpha_1 = \alpha_2 = \alpha_3 = 0$. The above is equivalent to

$$\underbrace{\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 5 \\ 3 & 0 & 9 \end{bmatrix}}_{\det = 0} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Indeed $3 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

($\alpha_1 = 3$, $\alpha_2 = -1$, $\alpha_3 = -1$), and thus the set is <u>linearly dependent.</u>

(b) $\mathbb{R}^3$ has dimension 3, and thus any set of 4 vectors must be <u>linearly dependent</u>. We do not need to do any further calculations.

(c). $$\alpha_1 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

if, and only if,

$$\underbrace{\begin{bmatrix} 3 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{\det = 1 \neq 0} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore$ Only solution is $\alpha_1 = \alpha_2 = \alpha_3 = 0$ and the vectors are <u>linearly independent</u>.

3. Claim: $\{v, w\}$ is linearly independent in $(V, \mathcal{F})$ if, and only if, $\{(v+w), (v-w)\}$ is linearly independent.

<u>Proof:</u>

a) Suppose $\{v, w\}$ is linearly indep. To show: $\{(v+w), (v-w)\}$ is linearly independent.

Consider $\quad \alpha(v+w) + \beta(v-w) = 0 \iff$

$$(\alpha+\beta)v + (\alpha-\beta)w = 0$$

By the linear independence of $\{v, w\}$, we have

$$\alpha + \beta = 0$$
$$\alpha - \beta = 0$$

$$\therefore \quad \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{\det \neq 0} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\therefore$ $\alpha = 0$ and $\beta = 0$ and then

$\{(v+w), (v-w)\}$ is linearly independent.

b) Suppose $\{(v+w), (v-w)\}$ is linearly independent. To show: $\{v, w\}$ is linearly independent.

Consider $\alpha v + \beta w = 0 \iff$

$$\frac{1}{2}(\alpha+\beta)(v+w) + \frac{1}{2}(\alpha-\beta)(v-w) = 0$$

By the linear independence of $\{(v+w), (v-w)\}$, this is possible if, and only if

$$\frac{\alpha+\beta}{2} = 0 \quad \text{and} \quad \frac{\alpha-\beta}{2} = 0$$

$$\iff \underbrace{\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{\det \neq 0} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \iff \alpha = 0 \text{ and } \beta = 0.$$

$\square$

4. Suppose that

$$\alpha \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} + \beta \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} + \gamma \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

This is equivalent to

$$\alpha + 2\beta + 4\gamma = 0$$

$$2\alpha + \beta - \gamma = 0$$

$$\alpha + \beta + \gamma = 0$$

(There are only 3 equations because the matrices are <u>symmetric</u>).

Hence

$$\underbrace{\begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}}_{\det = 0} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Hence, there will exist a nontrivial solution. Indeed, one solution is

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}.$$

Hence, the set is linearly dependent.

5. Claim. Let $Y \subset X$ be a subspace and $S \subset Y$. Then $\operatorname{span}\{S\} \subset Y$.

<u>Proof</u>: From the definition of span, if $S_1 \subset S_2$, then $\operatorname{span}\{S_1\} \subset \operatorname{span}\{S_2\}$. Also, if Y is a subspace, then $\operatorname{span}\{Y\} = Y$. Putting these two observations together we have

$$S \subset Y,\ Y \text{ a subspace} \Rightarrow \operatorname{span}\{S\} \subset Y.$$

$\square$

6.

(a) $\Rightarrow$ (b). Suppose $x \in V + W$ is written as

$$x = v_1 + w_2$$

and as

$$x = v_2 + w_2$$

where $v_1, v_2 \in V$ and $w_1, w_2 \in W$.

<u>To show:</u> $v_1 = v_2$ and $w_1 = w_2$

<u>Proof</u>

$$0 = x - x = (v_1 - v_2) + (w_1 - w_2)$$

Because V and W are subspaces,

$v_1 - v_2 \in V$ and $w_1 - w_2 \in W$. Because

$$v_1 - v_2 = w_2 - w_1 = -(w_1 - w_2)$$

we have $v_1 - v_2 \in W$ and $w_1 - w_2 \in V$.

Hence $(v_1 - v_2) \in V \cap W$ and $(w_1 - w_2) \in V \cap W$.

However, $V \cap W = \{0\}$, and thus $v_1 - v_2 = 0$ and $w_1 - w_2 = 0$, which is what we needed to show.

$\square$

In order to prove (b) $\Rightarrow$ (a), we will prove that $\sim(a) \Rightarrow \sim(b)$.

$\sim(a) \Leftrightarrow \exists\, u \in V \cap W$ with $u \neq 0$.

$\sim(b) \Leftrightarrow \exists\, x \in V + W$ that can be expressed in at least two different ways (i.e., NOT UNIQUE).

Because $x \in V + W$, $\exists\, v \in V$ and $w \in W$ such that $x = v + w$. We have that

$$x = v + w + u - u = (x + u) + (w - u)$$

also holds. Moreover, because $u \in V \cap W$, and because V and W are subspaces, we have that $v+u \in V$ and $w-u \in W$. Because $u \neq 0$, we have that

$$v \neq v+u \quad \text{and} \quad w \neq w-u.$$

So $\quad x = v+w$ and $x = (v+u)+(w-u)$ are distinct ways of expressing x as a sum of elements from V and W, and thus the representation is not unique.

7. $v^1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, v^2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}, v^3 = \begin{bmatrix} 2 \\ 8 \\ -4 \\ 8 \end{bmatrix}$

$v^4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, v^5 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ 6 \end{bmatrix}$

- $\{v^1, v^2\}$ linearly indep
- $v^3 = 4v^1 - 2v^2$
- $\{v^1, v^2, v^4\}$ linearly indep
- $v^5 = v^1 + v^2 + v^4$
- $\text{span}\{v^1, v^2, v^3, v^4, v^5\} = \text{span}\{v^1, v^2, v^4\}$

and $\{v^1, v^2, v^4\}$ is linearly independent.

$\therefore$ dimension is 3. $\square$

8 (b). For $1 \le i \le m$, and $1 \le j \le n$, Let E_{ij} have a 1 in the ij-entry and 0 elsewhere. These matrices are clearly a basis for $\mathbb{F}^{m,n}$. There are $m \times n$ such matrices.

$\therefore$ dimension $= m \times n$.

(c) Let E_{ij} be as above with $m=n$, and define $ES_{ij} = \frac{E_{ij} + E_{ji}}{2}$. For example,

$$ES_{11} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$ and

$$ES_{12} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & & \\ 0 & 0 & 0 & & \\ \vdots & & & \ddots & \\ \vdots & & & & \\ 0 & & & & 0 \end{bmatrix}$$

The set $\{ES_{ij} \mid 1 \le i \le n,\ i \le j \le n\}$ is a basis for the symmetric matrices.

The number of elements is $\frac{n(n+1)}{2}$.

$\square$